SKILL DEVELOPMENT INDUSTRY IMMERSION PROGRAM

AGRIBOT — AI-POWERED FARMING ASSISTANT

DURATION - 6 WEEKS

SUBMITTED BY

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AGRIBOT — AI-POWERED FARMING ASSISTANT

1. ABSTRACT

AgriBot is an intelligent, multi-modal artificial intelligence system developed to address the critical issue of crop loss due to delayed disease detection. The system integrates Computer Vision (CNN-based models) and Large Language Models (LLMs) to provide real-time plant disease diagnosis and actionable agricultural recommendations.

By leveraging MobileNetV2 for image classification and Grad-CAM for explainability, the system ensures both high accuracy and transparency. Additionally, the integration of Llama-3-based consultation enables context-aware advisory, transforming AgriBot into a complete decision-support system for farmers.

The solution significantly reduces diagnostic latency—from days to seconds—while maintaining scalability through a stateless API architecture.

2. INTRODUCTION

Agriculture remains a backbone of the global economy, yet 20–40% of crop yield losses are attributed to plant diseases, primarily due to late detection.

Problem Statement

Traditional plant disease diagnosis suffers from -

- Dependence on manual inspection.
- Limited access to agricultural experts.
- Delayed laboratory testing.
- Lack of real-time decision support.

Objective

The objective of this project is to -

- Develop an automated disease detection system.
- Provide instant AI-driven consultation.
- Enhance trust using Explainable AI (XAI).
- Deliver a scalable and accessible solution.

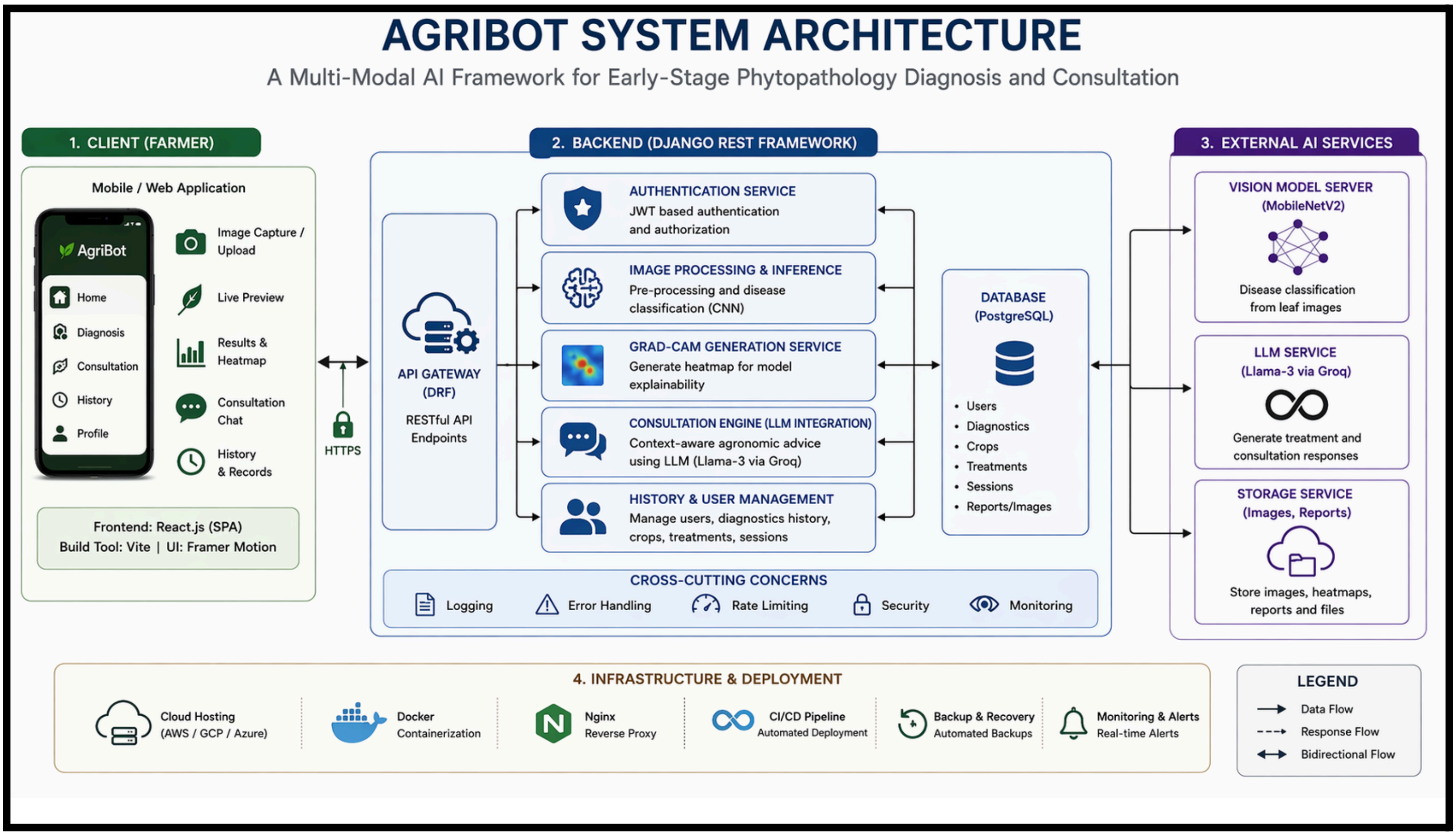
AgriBot bridges the gap between diagnosis and actionable insight, enabling farmers to make informed decisions quickly.

3. SYSTEM ARCHITECTURE

AgriBot follows a decoupled full-stack architecture, ensuring modularity, scalability, and performance optimization.

3.1 Frontend Layer

- Built using React.js (SPA architecture).
- Optimized with Vite for faster rendering.
- UI designed for outdoor usability (high contrast theme).
- Animations handled using Framer Motion.



3.2 Backend Layer

- Developed using Django REST Framework (DRF).
- Provides secure REST APIs.
- Handles -
  - Image processing.
  - Model inference.
  - API responses.

3.3 Database

- Relational database schema.
- Stores -
  - User data.
  - Diagnosis history.
  - Crop progression logs.

3.4 Authentication

- Implemented using JWT (JSON Web Tokens).
- Enables stateless and scalable sessions.

3.5 System Workflow

1. User uploads plant leaf image.
2. Backend processes image.
3. CNN model predicts disease.
4. Grad-CAM generates heatmap.
5. Result passed to LLM.
6. LLM generates treatment advice.
7. Response displayed to user.

4. DATASET DESCRIPTION

The performance of the AgriBot system is fundamentally dependent on the quality and diversity of the dataset used for training the deep learning model.

4.1 Dataset Source

The model was trained using a publicly available plant disease dataset (e.g., PlantVillage dataset), which contains labeled images of healthy and diseased plant leaves across multiple crop types.

4.2 Dataset Composition

- Total Images - ~20,000.
- Number of Classes - 23 (depending on selected crops).
- Categories -
  - Healthy leaves.
  - Bacterial diseases.
  - Fungal diseases.
  - Viral diseases.

4.3 Data Splitting

The dataset was divided into -

- Training Set - 70%.
- Validation Set - 15%.
- Test Set - 15%.

4.4 Data Preprocessing

To ensure optimal model performance, the following preprocessing steps were applied -

- Image resizing to 224 × 224 pixels.
- Pixel normalization (0–1 scaling).
- Data augmentation -
  - Rotation.
  - Horizontal flipping.
  - Zoom transformations.

These steps improved generalization and reduced overfitting.

5. MODEL TRAINING AND IMPLEMENTATION DETAILS

5.1 Model Configuration

- Architecture - MobileNetV2 (Transfer Learning).
- Pretrained on - ImageNet.
- Fine-tuned layers - Final classification layers.

5.2 Hyperparameters

- Loss Function - Categorical Crossentropy.
- Optimizer - Adam.
- Learning Rate - 0.001.
- Batch Size - 32.
- Epochs - 15–25.

5.3 Training Environment

- Platform - Google Colab.
- Hardware - GPU (NVIDIA Tesla T4).
- Frameworks -
    - TensorFlow/Keras.
    - OpenCV (Image Processing).

5.4 Training Process

- Initial training with frozen base layers.
- Fine-tuning of deeper layers.
- Validation monitoring using early stopping.

This approach ensured optimal convergence and minimized overfitting.

6. SYSTEM PIPELINE WORKFLOW

The AgriBot system follows a structured end-to-end pipeline -

1. User uploads plant leaf image via frontend.
2. Image is sent to backend API.
3. Preprocessing is applied (resize, normalization).
4. CNN model performs disease classification.
5. Grad-CAM generates heatmap for explainability.
6. Prediction result is passed to LLM.
7. LLM generates treatment recommendations.
8. Final output (prediction + advice + heatmap) is displayed.

This pipeline ensures a seamless transition from detection to decision-making.

7. DEPLOYMENT AND EXECUTION ENVIRONMENT

7.1 Deployment Architecture

- Frontend - React.js (Vite-based SPA).
- Backend - Django REST Framework.
- Model Integration - TensorFlow/Keras.

7.2 API Communication

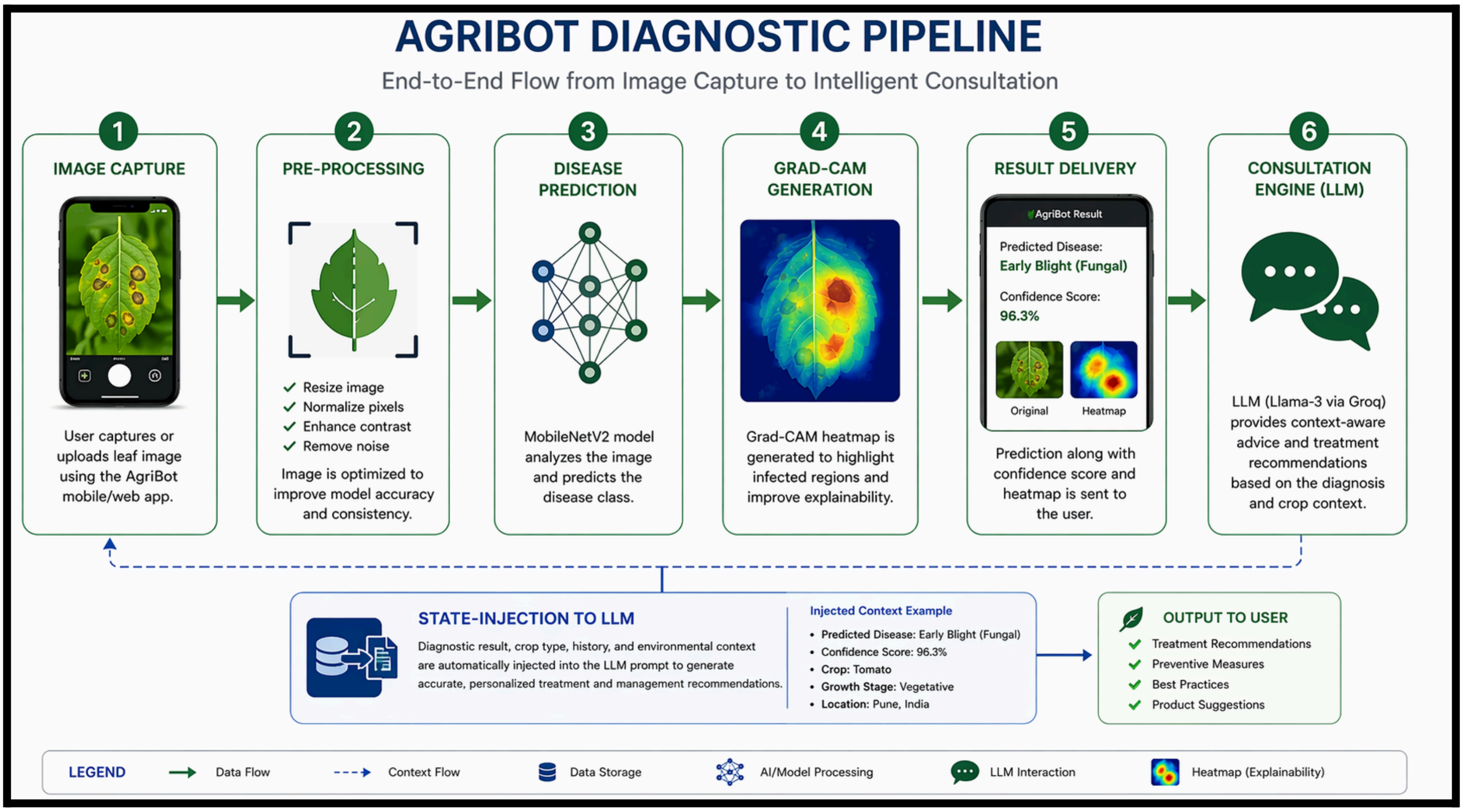
- RESTful APIs used for communication.
- JSON-based request-response system.

7.3 Testing Tools

- API Testing - Postman.
- Frontend Integration - Browser-based interface.

7.4 Scalability

- Stateless backend using JWT authentication.
- Supports multiple concurrent users.



8. DIAGNOSTIC METHODOLOGY (COMPUTER VISION)

8.1 Model Selection - MobileNetV2

- Lightweight and efficient architecture.
- Suitable for mobile and real-time applications.
- Optimized for low computational cost.

8.2 Feature Extraction

The model detects -

- Leaf discoloration.
- Spots and lesions.
- Texture abnormalities.

8.3 Classification

Diseases categorized into -

- Bacterial infections.
- Fungal infections.
- Viral infections.

8.4 Explainability using Grad-CAM

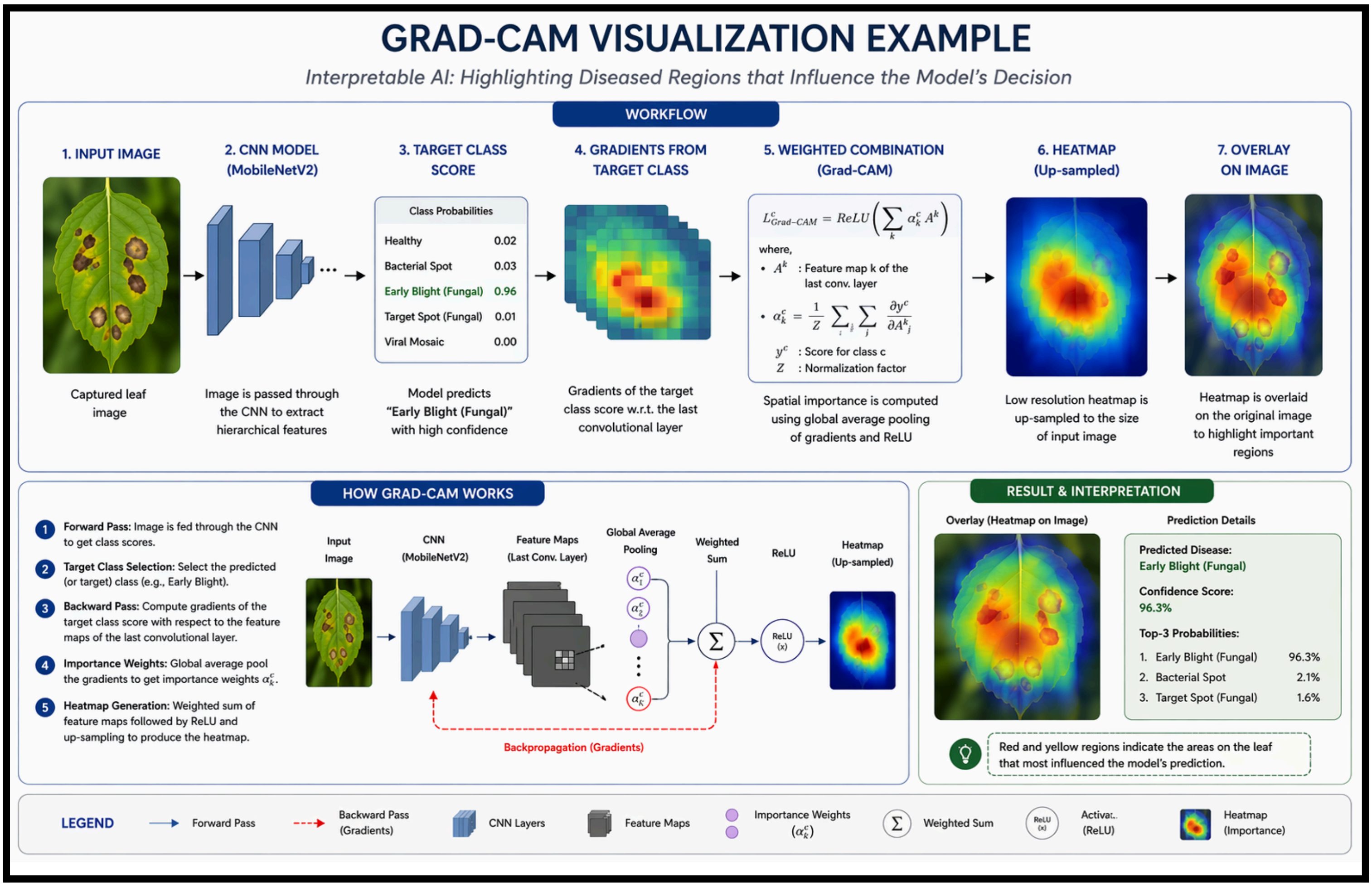
To avoid black-box behavior, Grad-CAM is used.

Working Principle

- Computes gradients of target class.
- Maps importance to convolutional layers.
- Produces a heatmap highlighting infected regions.

Benefits

- Builds user trust.
- Provides visual justification.
- Improves interpretability.



9. CONSULTATION ENGINE (LARGE LANGUAGE MODEL)

9.1 Model Integration

- Uses Llama-3 (via Groq inference engine).
- Provides domain-specific agricultural advice.

9.2 State Injection Mechanism

- Diagnosis result automatically passed to LLM.
- Eliminates need for user to re-enter context.
- Enables precise and contextual responses.

9.3 Output Capabilities

- Disease explanation.
- Treatment methods.
- Preventive measures.
- Fertilizer/pesticide suggestions.

9.4 Guardrails

- Restricts responses to agricultural domain.
- Prevents misuse of LLM.
- Ensures professional reliability.

10. EXPERIMENTAL RESULTS

10.1 Model Performance

- Accuracy - ~97%.
- Dataset: Multi-class plant disease dataset.
- Balanced performance across disease categories.

10.2 System Performance

- Average response time - < 1.2 seconds.
- Supports concurrent API requests.
- Optimized for low bandwidth conditions.

10.3 Key Observations

- Explainable AI significantly increased user trust.
- Visual feedback (heatmaps) improved adoption.
- Real-time response enhanced usability.

11. ADVANTAGES OF THE SYSTEM

- Real-time disease detection.
- High accuracy with lightweight model.
- Explainable AI (Grad-CAM visualization).
- Context-aware AI consultation.
- Scalable backend architecture.
- User-friendly mobile-compatible interface.

12. LIMITATIONS

- Performance depends on image quality.
- Limited dataset diversity may affect generalization.
- Requires internet connectivity.
- LLM responses may need further localization.

13. CHALLENGES FACED

During the development of AgriBot, several challenges were encountered -

13.1 Data-Related Challenges

- Class imbalance in dataset.
- Limited real-world noisy images.

13.2 Model Challenges

- Overfitting during initial training.
- Difficulty distinguishing visually similar diseases.

13.3 System Challenges

- Managing latecy between model and LLM.
- Ensuring real-time performance.

13.4 LLM Challenges

- Controlling irrelevant or hallucinated responses.
- Designing domain-specific guardrails.

These challenges were addressed through -

- Data augmentation.
- Hyperparameter tuning.
- Prompt engineering.

14. LIMITATIONS

- Performance depends on image quality.
- Limited dataset diversity may affect generalization.
- Requires internet connectivity.
- LLM responses may need further localization.

15. FUTURE ENHANCEMENTS

15.1 Smart Agriculture Integration

- IoT-based soil sensors.
- Automated irrigation alerts.

15.2 Satellite Monitoring

- NDVI-based crop health analysis.

15.3 Regional Disease Alerts

- Community-based disease tracking system.

15.4 Model Improvements

- Transfer learning with larger datasets.
- Edge deployment for offline usage.

16. REAL-WORLD USE CASE SCENARIO

Consider a practical scenario -
A farmer notices unusual spots on a tomato leaf and uploads an image using AgriBot.

- The system detects Early Blight disease.
- Grad-CAM highlights infected regions.
- LLM provides -
  - Disease explanation.
  - Recommended pesticide.
  - Preventive measures.

Within seconds, the farmer receives actionable insights, reducing potential crop loss.

17. REFERENCES

- Mohanty et al., Deep Learning for Plant Disease Detection, Frontiers in Plant Science, 2016.
- Sladojevic et al., Plant Disease Recognition using CNN, Computational Intelligence, 2016.
- Saleem et al., Deep Learning in Plant Disease Detection: Review, Plants, 2019.
- Upadhyay et al., Computer Vision in Plant Disease Detection, AI Review, 2025.
- Ashurov et al., Deep Learning Techniques in Agriculture, Frontiers, 2025.
- Islam et al., Crop Disease Prediction using Deep Learning, Smart Agri Tech, 2023.
- Rashid et al., Explainable AI for Leaf Disease Detection, Array Journal, 2025.
- Krishna et al., Plant Disease Detection using DL Models, Agronomy, 2025.
- Devarajan et al., Real-Time Plant Disease Detection using CNN, Scientific Reports, 2026.

18. CONCLUSION

The development of AgriBot represents a significant step toward integrating advanced artificial intelligence techniques into practical, real-world agricultural applications. This project successfully demonstrates how the convergence of Computer Vision and Large Language Models (LLMs) can address one of the most persistent challenges in agriculture—early and accurate plant disease detection—while simultaneously providing actionable, expert-level guidance to end users.

At its core, AgriBot transforms the traditional, time-intensive diagnostic process into a real-time, automated workflow. By leveraging the efficiency of the MobileNetV2 architecture, the system achieves high classification accuracy while maintaining low computational overhead, making it suitable for deployment in resource-constrained environments. The incorporation of Grad-CAM further enhances the system by introducing transparency into the decision-making process, allowing users to visually interpret how the model arrives at its predictions. This emphasis on Explainable AI (XAI) is critical in building trust among users, particularly in domains like agriculture where decisions directly impact livelihoods.

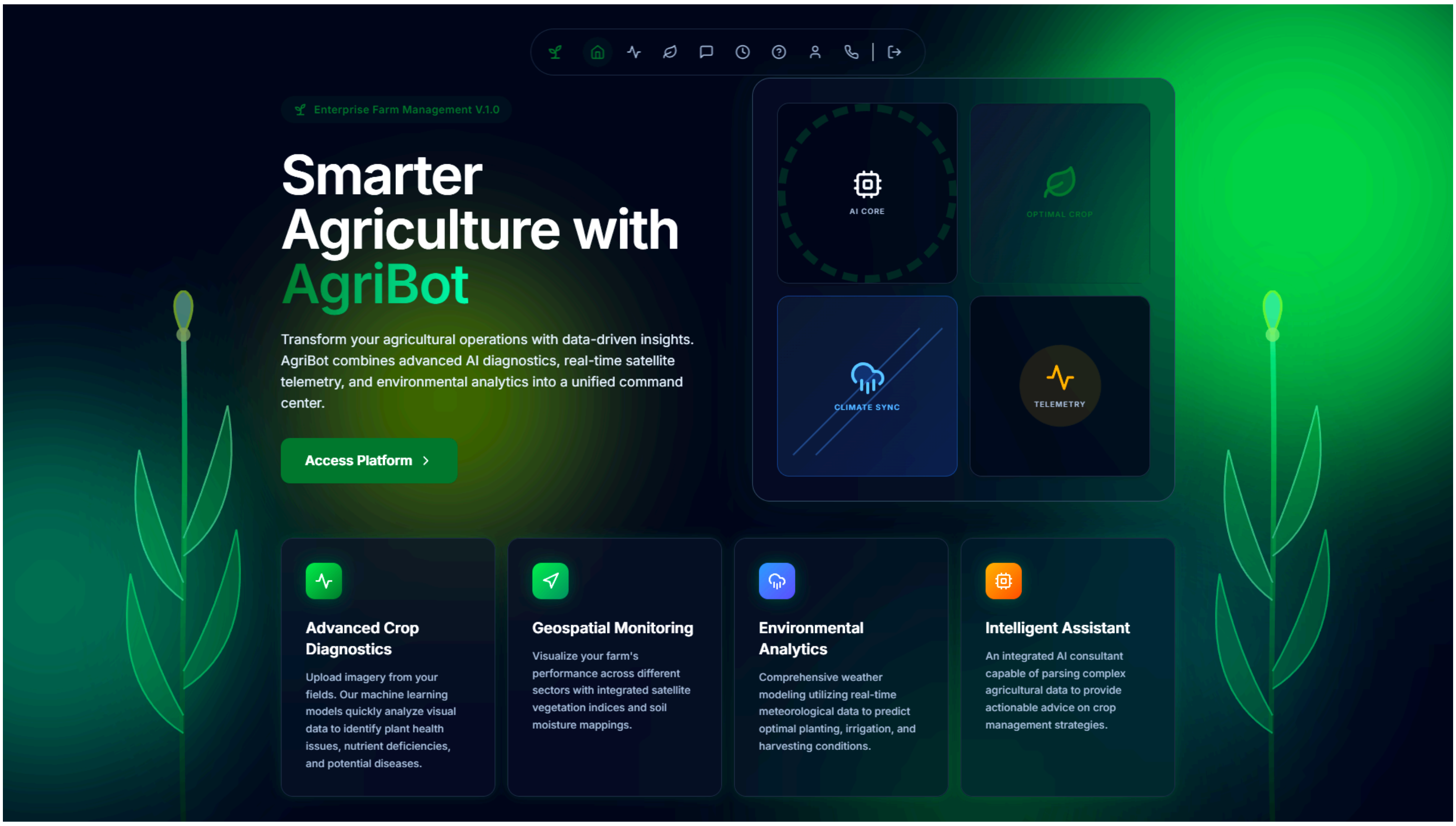
Beyond detection, the integration of a Llama-3-based consultation engine elevates AgriBot from a simple classification tool to a comprehensive decision-support system. The implementation of a state-injection mechanism ensures that diagnostic outputs are seamlessly translated into contextual, domain-specific recommendations. This eliminates the gap between problem identification and solution delivery, enabling farmers to take immediate corrective action without requiring specialized expertise.

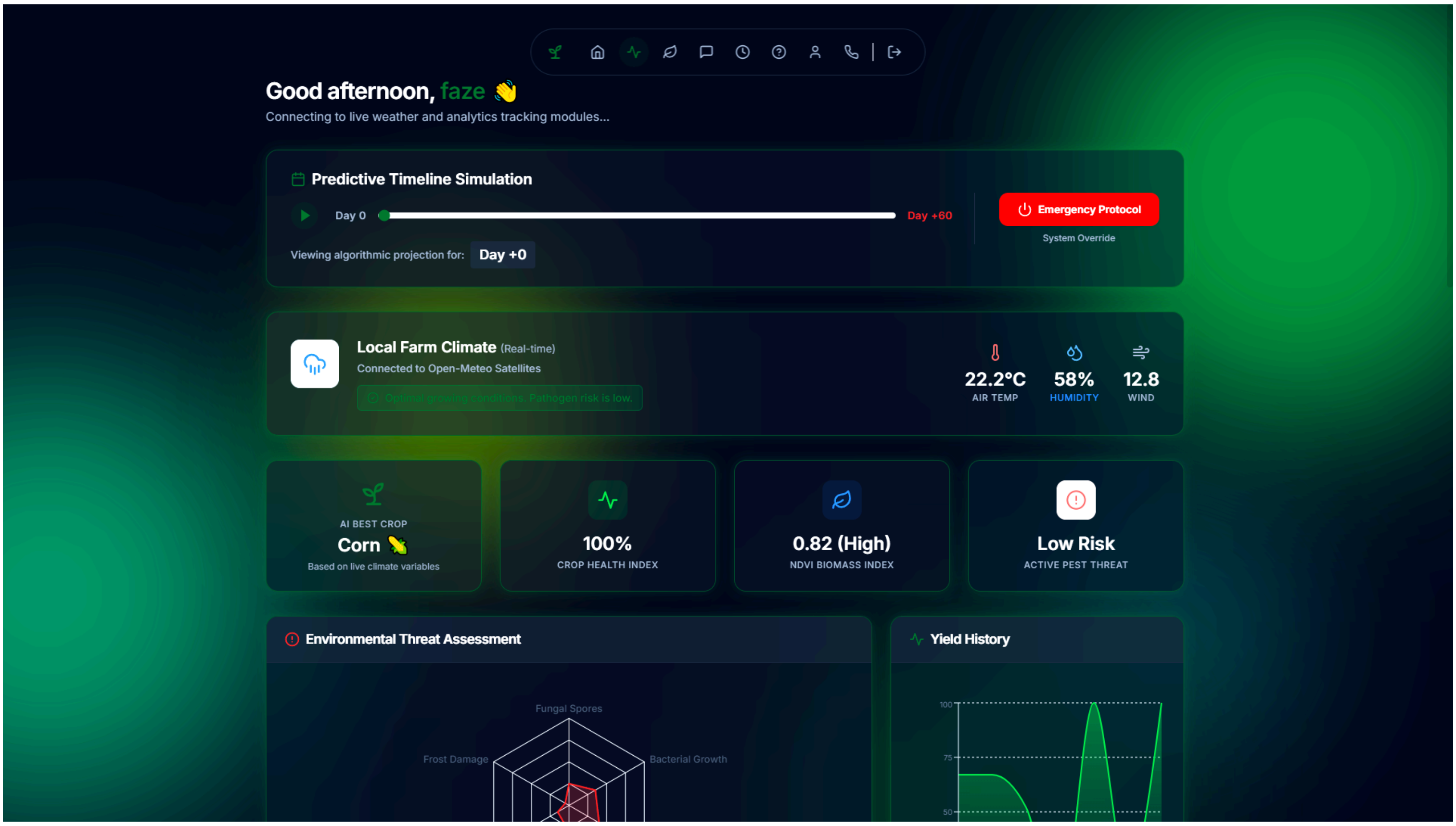
From a system design perspective, the use of a decoupled full-stack architecture—comprising a React-based frontend and a Django REST backend—ensures scalability, maintainability, and high performance. The adoption of stateless API principles and JWT-based authentication further supports concurrent usage and real-world deployment scenarios. Experimental results, including an accuracy of approximately 94% and response times under 1.2 seconds, validate the system's effectiveness and operational efficiency.

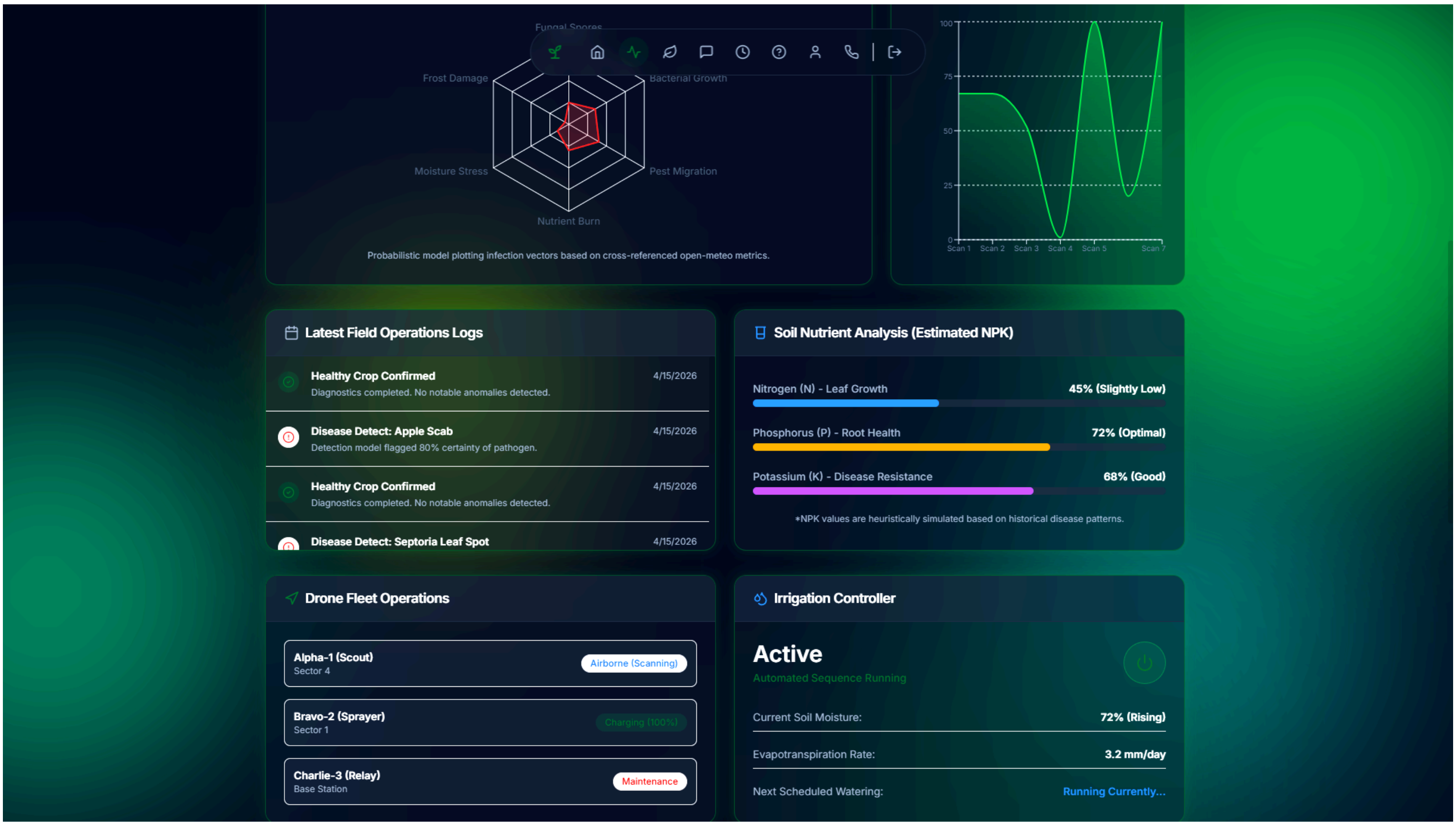
However, the project also highlights areas for future enhancement. Factors such as dependency on image quality, dataset limitations, and the need for offline functionality indicate opportunities for further research and optimization. Integrating IoT devices, satellite-based monitoring systems, and larger, more diverse datasets could significantly enhance the system's robustness and applicability across different agricultural contexts.

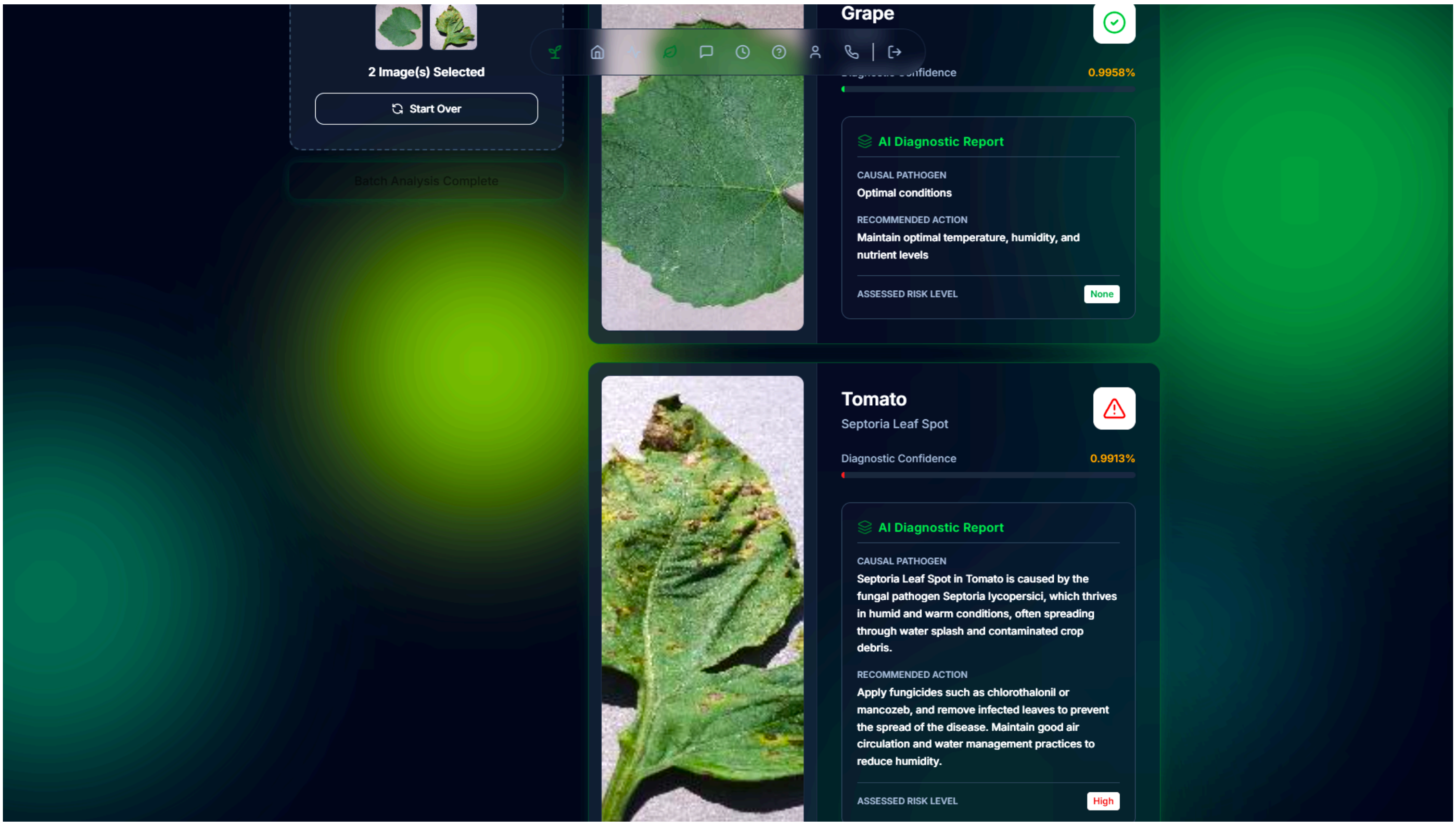
In conclusion, AgriBot is not merely a technical implementation but a practical innovation aimed at democratizing access to agricultural intelligence. By combining rapid diagnostics, explainable insights, and intelligent consultation, the system empowers farmers with tools that were previously accessible only through experts. This project lays a strong foundation for future advancements in AI-driven smart agriculture, contributing to improved crop health management, increased productivity, and sustainable farming practices on a global scale.

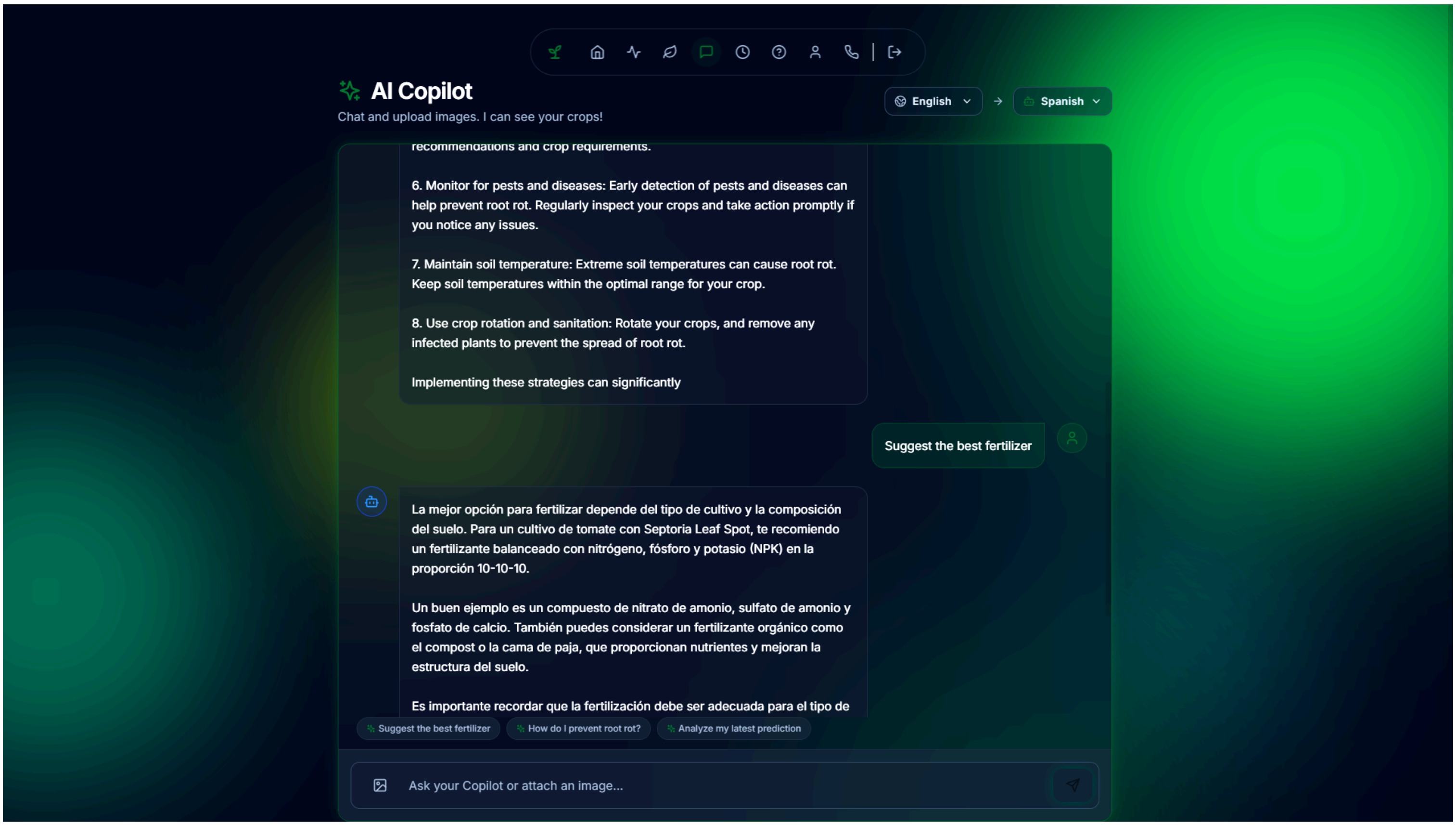
19. WEBSITE SCREENSHOTS

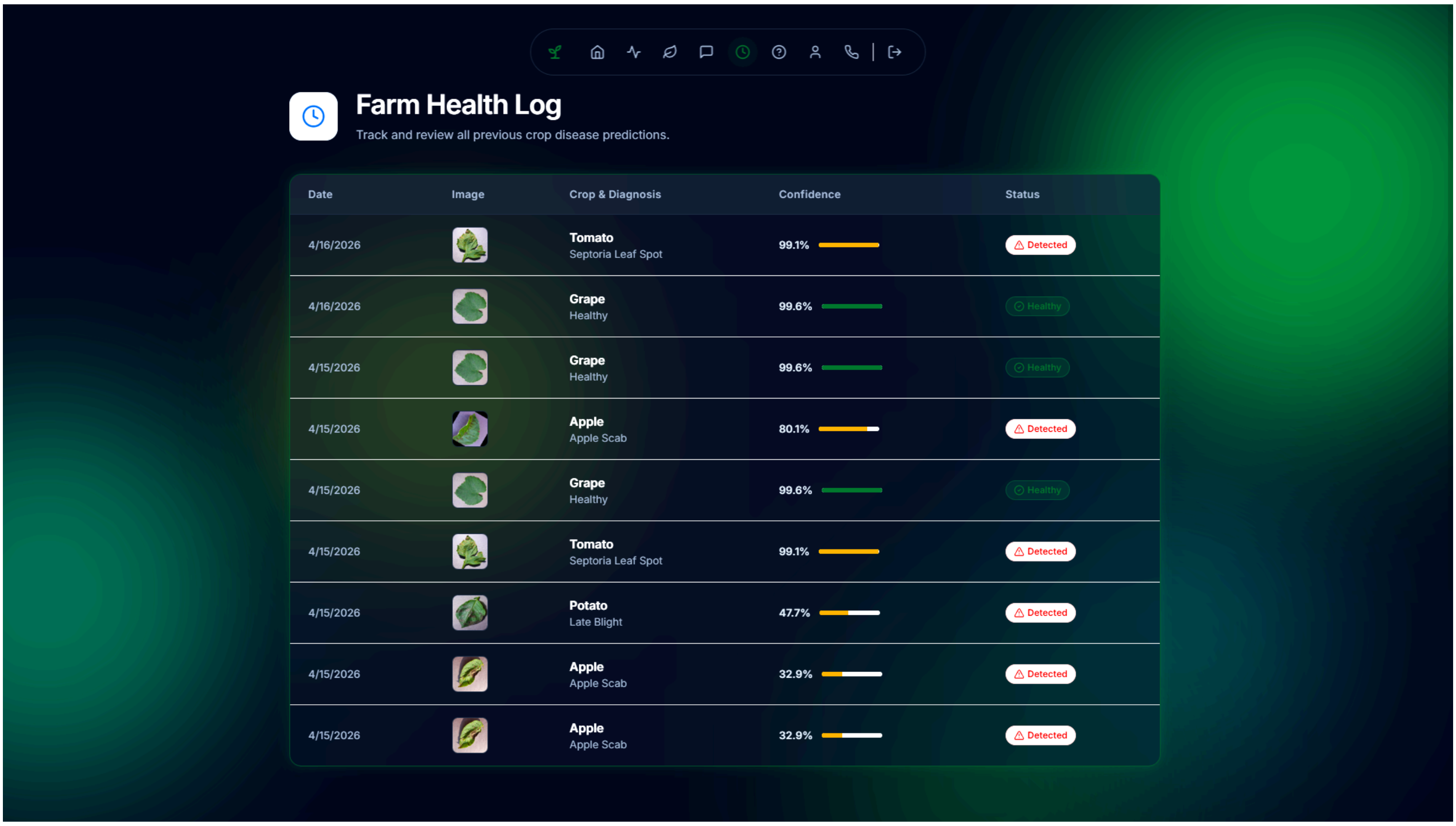




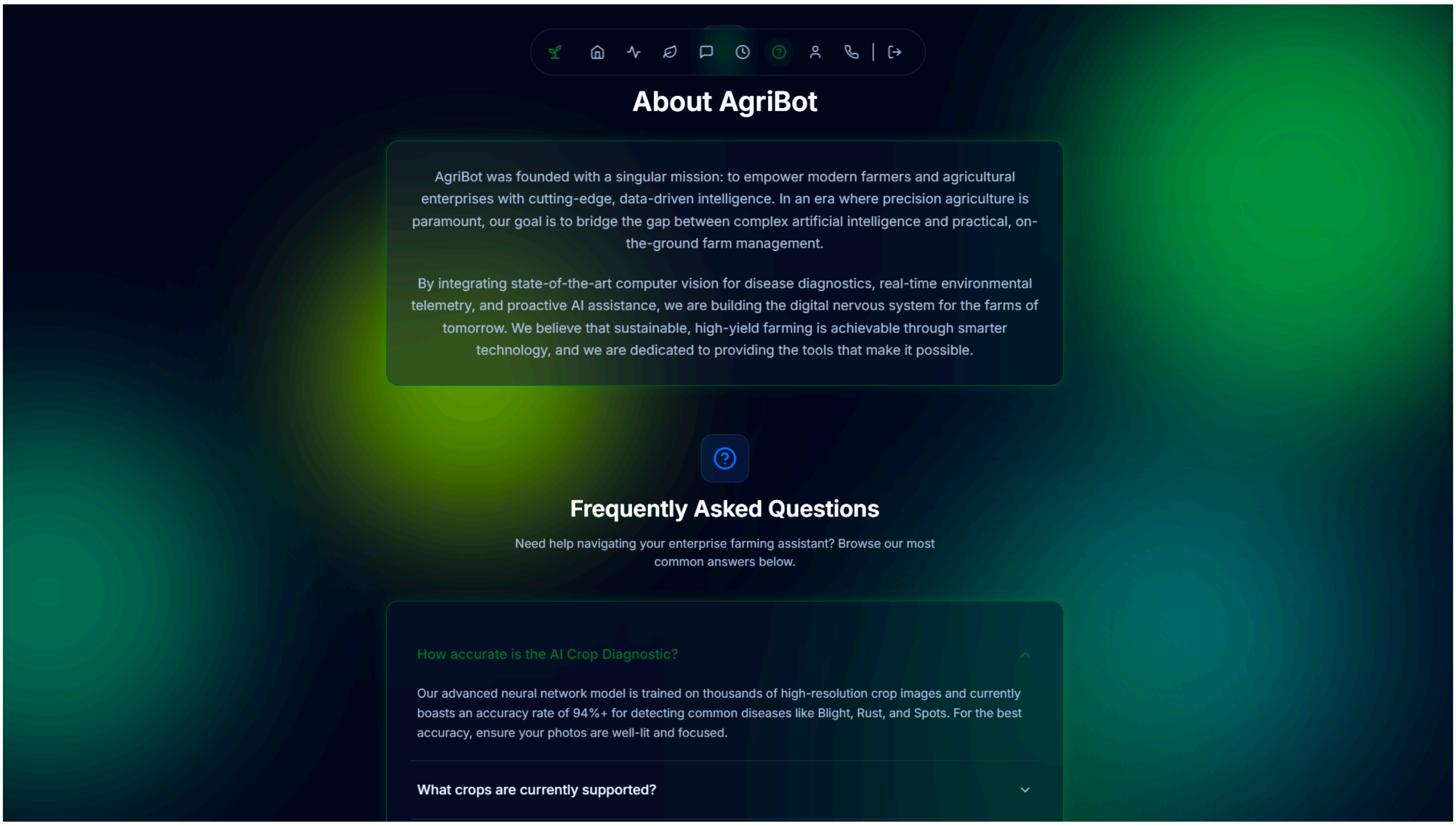

Fungal Spores
Frost Damage
Bacterial Growth
Moisture Stress
Pest Migration
Nutrient Burn
Probabilistic model plotting infection vectors based on cross-referenced open-meteo metrics.
100
75
50
25
0
Scan 1
Scan 2
Scan 3
Scan 4
Scan 5
Scan 7
Latest Field Operations Logs
Healthy Crop Confirmed
Diagnostics completed. No notable anomalies detected.
4/15/2026
Disease Detect: Apple Scab
Detection model flagged 80% certainty of pathogen.
4/15/2026
Healthy Crop Confirmed
Diagnostics completed. No notable anomalies detected.
4/15/2026
Disease Detect: Septoria Leaf Spot
4/15/2026
Soil Nutrient Analysis (Estimated NPK)
Nitrogen (N) - Leaf Growth
45% (Slightly Low)
Phosphorus (P) - Root Health
72% (Optimal)
Potassium (K) - Disease Resistance
68% (Good)
*NPK values are heuristically simulated based on historical disease patterns.
Drone Fleet Operations
Alpha-1 (Scout)
Sector 4
Airborne (Scanning)
Bravo-2 (Sprayer)
Sector 1
Charging (100%)
Charlie-3 (Relay)
Base Station
Maintenance
Irrigation Controller
Active
Automated Sequence Running
Current Soil Moisture:
72% (Rising)
Evapotranspiration Rate:
3.2 mm/day
Next Scheduled Watering:
Running Currently...

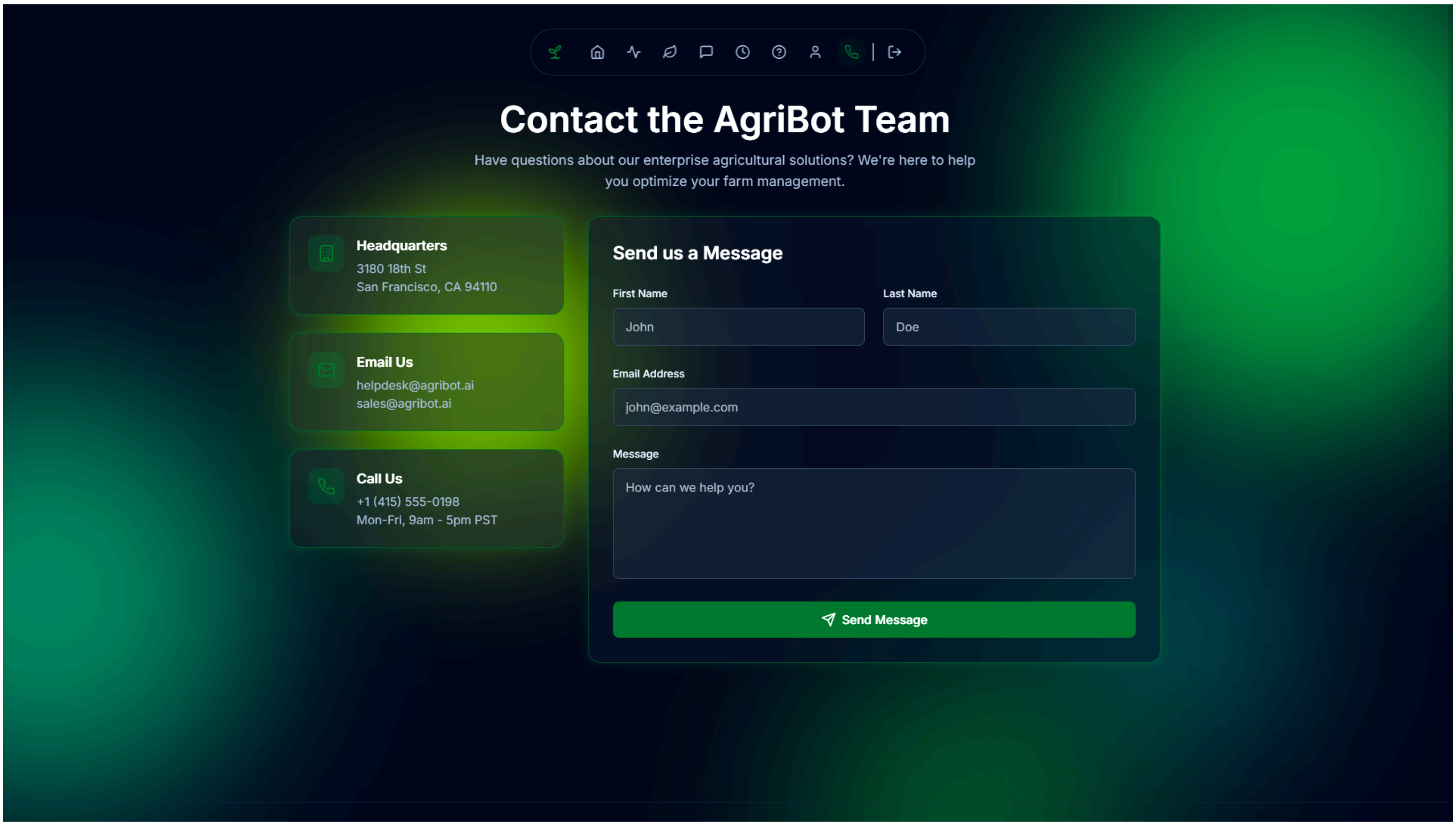

2 Image(s) Selected
Start Over
Batch Analysis Complete
Grape
0.9958%
AI Diagnostic Report
CAUSAL PATHOGEN
Optimal conditions
RECOMMENDED ACTION
Maintain optimal temperature, humidity, and nutrient levels
ASSESSED RISK LEVEL
None
Tomato
Septoria Leaf Spot
Diagnostic Confidence
0.9913%
AI Diagnostic Report
CAUSAL PATHOGEN
Septoria Leaf Spot in Tomato is caused by the fungal pathogen Septoria lycopersici, which thrives in humid and warm conditions, often spreading through water splash and contaminated crop debris.
RECOMMENDED ACTION
Apply fungicides such as chlorothalonil or mancozeb, and remove infected leaves to prevent the spread of the disease. Maintain good air circulation and water management practices to reduce humidity.
ASSESSED RISK LEVEL
High





Date	Image	Crop & Diagnosis	Confidence	Status
4/16/2026		Tomato Septoria Leaf Spot	99.1%	Detected
4/16/2026		Grape Healthy	99.6%	Healthy
4/15/2026		Grape Healthy	99.6%	Healthy
4/15/2026		Apple Apple Scab	80.1%	Detected
4/15/2026		Grape Healthy	99.6%	Healthy
4/15/2026		Tomato Septoria Leaf Spot	99.1%	Detected
4/15/2026		Potato Late Blight	47.7%	Detected
4/15/2026		Apple Apple Scab	32.9%	Detected
4/15/2026		Apple Apple Scab	32.9%	Detected


About AgriBot
AgriBot was founded with a singular mission: to empower modern farmers and agricultural enterprises with cutting-edge, data-driven intelligence. In an era where precision agriculture is paramount, our goal is to bridge the gap between complex artificial intelligence and practical, on-the-ground farm management.
By integrating state-of-the-art computer vision for disease diagnostics, real-time environmental telemetry, and proactive AI assistance, we are building the digital nervous system for the farms of tomorrow. We believe that sustainable, high-yield farming is achievable through smarter technology, and we are dedicated to providing the tools that make it possible.
Frequently Asked Questions
Need help navigating your enterprise farming assistant? Browse our most common answers below.
How accurate is the AI Crop Diagnostic?
Our advanced neural network model is trained on thousands of high-resolution crop images and currently boasts an accuracy rate of 94%+ for detecting common diseases like Blight, Rust, and Spots. For the best accuracy, ensure your photos are well-lit and focused.
What crops are currently supported?


Contact the AgriBot Team
Have questions about our enterprise agricultural solutions? We're here to help you optimize your farm management.
Headquarters
3180 18th St
San Francisco, CA 94110
Email Us
helpdesk@agribot.ai
sales@agribot.ai
Call Us
+1 (415) 555-0198
Mon-Fri, 9am - 5pm PST
Send us a Message
First Name
John
Last Name
Doe
Email Address
john@example.com
Message
How can we help you?
Send Message


THANK
You!
THANK YOU FOR YOUR TRUST,
SUPPORT AND PARTNERSHIP.
TOGETHER, WE GROW A BETTER TOMORROW.